1.7 De Morgan's Laws

Augustus De Morgan (1806 – 1871) was a British mathematician. He was born on 27th June 1806 in Madurai, Tamilnadu, India. His father was posted in India by the East India Company. When he was seven months old, his family moved back to England. De Morgan was educated at Trinity College, Cambridge, London. He formulated laws for set difference and complementation. These are called De Morgan's laws.

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1.7.1 De Morgan's Laws for Set Difference

These laws relate the set operations union, intersection and set difference.

Let us consider three sets A, B and C as $A = \{-5, -2, 1, 3\}$, $B = \{-3, -2, 0, 3, 5\}$ and $C = \{-2, -1, 0, 4, 5\}$.

Now, $B \cup C = \{-3, -2, -1, 0, 3, 4, 5\}$

$A - (B \cup C) = \{-5, 1\}$ ... (1)

Then, $A - B = \{-5, 1\}$ and $A - C = \{-5, 1, 3\}$

$(A - B) \cup (A - C) = \{-5, 1, 3\}$ ... (2)

$(A - B) \cap (A - C) = \{-5, 1\}$ ... (3)

From (1) and (2), we see that

$A - (B \cup C) \neq (A - B) \cup (A - C)$

But note that from (1) and (3), we see that

$A - (B \cup C) = (A - B) \cap (A - C)$

Thinking Corner

$(A - B) \cup (A - C) \cup (A \cap B) =$ ____

Now, $B \cap C = \{-2, 0, 5\}$

$A - (B \cap C) = \{-5, 1, 3\}$ ... (4)

From (3) and (4) we see that

$A - (B \cap C) \neq (A - B) \cap (A - C)$

But note that from (2) and (4), we get $A - (B \cap C) = (A - B) \cup (A - C)$

De Morgan's laws for set difference : For any three sets A, B and C

(i) $A - (B \cup C) = (A - B) \cap (A - C)$ (ii) $A - (B \cap C) = (A - B) \cup (A - C)$

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